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and multinucleate structures which bear bag like structure called sporangia. Each sporangiophore bears a single sporangium.

Sporangium possesses a sterile region in the centre called **Columella**. Spores are produced around the columella. When the sporangial wall breaks, the columella collapses and the spores are dispersed. When the spores fall on a suitable substratum they germinate and produce new mycelia (Figure 1.26).

Sexual reproduction

Sexual reproduction is present and takes place through gametangial copulation. Most of the species are heterothallic but *Rhizopus sexualis* is homothallic. There is no morphological distinction between the two sexual hyphae although physiologically they are dissimilar. Since physiologically dissimilar thalli (hyphae) are involved in sexual reproduction, this phenomenon is called **heterothallism**. Mycelia which produce gametangia are of opposite strains (+) or (-). The first step is the formation of special hyphae called zygophores. The tips of the two zygophores swell to form progametangia. Further, a septum is formed near the tip of each progametangium and results in the formation of a terminal gametangium and a suspensor cell. The two gametangia fuse, and this is followed by plasmogamy and karyogamy. The fusion of nuclei results in the formation of a diploid zygospore. Many nuclei belonging to opposite strains (+ or –) pair and fuse to form many diploid nuclei. The zygospore enlarges and develops an outer thick dark and warty layer called exine and inner thin layer called intine. After the resting period the nuclei of zygospore undergo meiosis. The zygospore germinates to form sporangiophores and the zygosporangium contain mixture of (+) and (–) spores. When the spores fall on a suitable substratum, they germinate to produce mycelium (Figure 1.20). The life cycle of *Rhizopus* is given in figure 1.27.

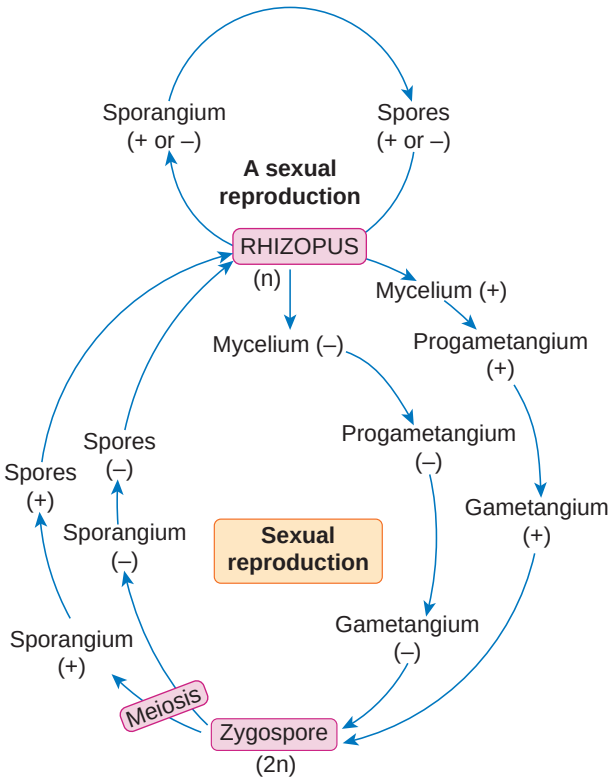


Figure 1.27: Life cycle of *Rhizopus*

1.5.8 Agaricus

Class - Basidiomycetes
Order - Agaricales
Family - Agaricaceae
Genus - *Agaricus*

It is a saprophytic fungus found on wood logs, manure piles, fresh litter, pastures etc., The fruit bodies are the visible part of the fungi. They are found in rings in some species like *Agaricus arvensis*, *Agaricus tabularis* and hence popularly called 'Fairy rings". *Agaricus campestris* is the most common 'field mushroom'.

Vegetative structure

The thallus is made up of branched structures called hyphae. A large number of hyphae constitute the mycelium.